Predicting Road Collision Severity in London - Integrating Network Centrality, Traffic Speed, and Deprivation (2020-2024)

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Source code and data can be found here.

HTML version can be found here.

1. Introduction

Road traffic collisions are a leading cause of preventable death and serious injury. In London, despite a Vision Zero strategy targeting a 70% reduction in Killed or Seriously Injured (KSI) casualties by 2030, approximately 3,700 people were killed or seriously injured on roads in 2024, with nearly 60% comprising pedestrians and cyclists (TfL, 2024). KSI, defined within STATS19 as fatalities and injuries involving fracture, concussion, internal trauma, or inpatient hospitalisation, represents an area for critical attention. Understanding *which* collisions result in KSI outcomes and *why* has implications for infrastructure investment, emergency resource allocation, and targeted safety intervention.

KSI risk determinants are multi-dimensional. Speed limit, lighting, road surface, and junction characteristics emerge as primary severity predictors in ensemble modelling studies (Tong, 2025). Deprivation compounds this, with pedestrians in the most deprived neighbourhoods facing higher KSI probability than those in least deprived areas (Christie et al., 2019).

Critically, Sarkar, Webster & Kumari (2018) demonstrated in a Greater London study that street network topology, specifically betweenness centrality, divergence ratio, and hull radius, were associated with KSI odds alongside speed and deprivation, establishing that a location's structural position within the road network shapes severity outcomes.

This study combines OpenStreetMap-derived centrality measures with STATS19, the Index of Multiple Deprivation, DfT AADF traffic counts, and LAEI 2022 road speed data to predict KSI outcomes across London (2020–2024). Three tree-based classifiers are evaluated under stratified cross-validation with Synthetic Minority Over-sampling Technique (SMOTE) (Chawla et al., 2002), and model decisions interpreted with SHAP (Lundberg & Lee, 2017) to surface key predictors of collision severity.

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2. Research Question

Primary Research Question: To what extent do graph-derived network centrality measures, deprivation indicators, and casualty-level vulnerability factors jointly predict KSI outcomes in London road collisions (2020–2024)?

Sub-questions:

- **Network structure:** Do closeness and betweenness centrality independently predict elevated KSI risk after controlling for traffic volume, speed, and road environment, and does including graph features measurably improve model performance?
- **Deprivation:** Which aspects of deprivation (area-level, driver, or casualty) most strongly shape KSI severity outcomes?
- **Casualty vulnerability:** How do casualty characteristics such as mode of travel (pedestrian, cyclist, motorcyclist), age group, and collision complexity shape KSI probability?

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3. Data

3.1 Data Sources

The analytical feature matrix was constructed from seven datasets.

Dataset	Source	Spatial Coverage	Temporal Coverage
STATS19 — Collisions	DfT / data.gov.uk	England & Wales	2020–2024
STATS19 — Vehicles	DfT / data.gov.uk	England & Wales	2020–2024
STATS19 — Casualties	DfT / data.gov.uk	England & Wales	2020–2024
London Road Network	OpenStreetMap via OSMnx	Greater London	Current
Index of Multiple Deprivation 2019	MHCLG / data.gov.uk	England (LSOA)	2019
Traffic Count Points & AADF	DfT / roadtraffic.dft.gov.uk	Great Britain	2000–2024
LAEI 2022 — Road Speeds (Major & Minor)	GLA / data.london.gov.uk	Greater London	2022

3.2 Data Preparation

Data was prepared using the following methods:

Category	Source / Method	Description & Preprocessing
Target Variable (KSI)	STATS19 (Greater London)	Binary classification: Severity 1 (Fatal) & 2 (Serious) aggregated into positive class; Severity 3 (Slight) as negative class.
Network Measures	Graph-based Centrality	Degree, betweenness, and closeness centrality; spatially snapped to nearest network node and joined to collision records.
Traffic Volume	DfT AADF (2020–2024)	Collisions were matched to nearest DfT traffic count point where total motor vehicle flow was log-transformed to normalize right-skewed distribution.
Road Speed Proxy	LAEI 2022 & Temporal Features	Proxy for impact speed using average operating speeds; integrated with diurnal features (hour/time-period). Although restricted-access OS NGD Transport Network data offers higher-resolution link-level speeds, the LAEI-temporal approach provides a minimal accessible alternative for capturing impact-related risk
Deprivation	IMD 2019 Deciles	Three-dimensional socioeconomic context: `imd_decile` (location), `mean_driver_imd`, and `mean_casualty_imd`.

3.3 Variables Selected

The final analytical feature set comprised 43 variables in 7 groups.

Target variable

Variable	Type	Definition
`severity_binary`	Binary	1 = KSI (Fatal or Serious); 0 = Slight. Derived from STATS19 `collision_severity` codes 1 and 2.

Graph centrality (3)

Variable	Type	Definition
`closeness_centrality`	Continuous	Reciprocal of mean shortest-path distance from node to all others; reflects network accessibility.
`betweenness_centrality`	Continuous	Proportion of all shortest paths passing through a node; reflects throughput load and conflict potential.
`degree_centrality`	Continuous	Number of edges connected to a node, normalised by maximum possible; reflects junction complexity.

Temporal (4)

Variable	Type	Definition
`hour_of_collision`	Integer	Hour of day (0–23); proxy for prevailing operating speeds and traffic state.
`time_period_enc`	Ordinal	AM peak / inter-peak / PM peak / night; captures systematic temporal severity patterns.
`year`	Integer	Collision year (2020–2024); captures long-term drift and structural changes in road safety.
`quarter`	Ordinal	Calendar quarter (1–4); groups Jan-Mar, Apr-Jun, Jul-Sep, Oct-Dec to capture seasonal risk profiles.

Collision environment (9)

Variable	Type	Definition
`speed_limit`	Ordinal	Posted speed limit in mph (20–70).
`road_type`	Categorical	Roundabout, One-way street, single carriageway, dual carriageway, slip road
`first_road_class`	Categorical	Motorway, Major A-road, A-road, B-road, unclassified
`junction_detail`	Categorical	Not at junction or within 20 meters, T or staggered, crossroads, 4 arm junction
`junction_control`	Categorical	Traffic signals, give way, stop, uncontrolled.
`light_conditions`	Categorical	Daylight, darkness (lit / unlit / unknown lighting).
`weather_conditions`	Categorical	Fine, rain, snow, fog, high winds, etc.
`road_surface_conditions`	Categorical	Dry, wet, frost/ice, flood.
`pedestrian_crossing`	Categorical	Crossing facility type within 50 m of collision.

Deprivation — three dimensions (3)

Variable	Type	Definition
`imd_decile`	Ordinal	IMD 2019 decile of accident LSOA (1 = most deprived); captures road environment quality.
`mean_driver_imd`	Continuous	Mean IMD decile of drivers' home postcodes per collision; proxy for vehicle quality and risk behaviour.
`mean_casualty_imd`	Continuous	Mean IMD decile of casualties' home postcodes per collision; proxy for individual vulnerability.

Vehicle-level aggregates (12)

Variable	Type	Definition
`motorcycle_involved`	Binary	Any motorcycle or moped involved.
`hgv_involved`	Binary	Any heavy goods vehicle involved.
`cycle_involved`	Binary	Any pedal cycle involved.
`any_skid_overturn`	Binary	Any vehicle skidded or overturned.
`any_hit_offroad_object`	Binary	Any vehicle struck an off-road object (roadside barrier, tree, sign, etc.).
`any_left_carriageway`	Binary	Any vehicle left the carriageway.
`any_front_impact`	Binary	Any vehicle had a front-impact collision (head-on or rear-end).
`any_moving_vehicle`	Binary	At least one vehicle was moving at point of impact (vs. stationary).
`any_turning_vehicle`	Binary	At least one vehicle was executing a turn at point of impact.
`work_journey`	Binary	Any driver making a work or business journey at time of collision.
`mean_driver_age`	Continuous	Mean age of all drivers involved per collision.
`any_male_driver`	Binary	Any male driver involved; used as a risk behaviour proxy.

Casualty-level aggregates (9)

Variable	Type	Definition
`pedestrian_casualty`	Binary	At least one pedestrian casualty.
`motorcyclist_casualty`	Binary	At least one motorcyclist casualty.
`cyclist_casualty`	Binary	At least one cyclist casualty.
`child_casualty`	Binary	At least one casualty aged under 16.
`elderly_casualty`	Binary	At least one casualty aged 70 or over.
`min_casualty_age`	Integer	Age of youngest casualty; captures child involvement continuously.
`ped_on_crossing`	Binary	Pedestrian struck while on or near a marked crossing.
`ped_jaywalking`	Binary	Pedestrian struck along carriageway or crossing elsewhere
`n_casualties`	Integer	Total number of casualties per collision; measures collision severity scale.

Traffic volume and speed (3)

Variable	Type	Definition
`aadf_log`	Continuous	Log-transformed total annual average daily motor vehicle flow at nearest count point.
`accident_rate_clipped`	Continuous	Accidents per million vehicle-days at nearest count point (99th-percentile clipped).
`avg_speed_kph`	Continuous	LAEI 2022 average operating speed: major-road link speed (≤30 m match) or minor-road 1 km^2 grid fallback.

3.4 Data Quality

The table summarises data completeness at each integration stage.

Stage	Records	Notes
STATS19 national (2020–2024)	~1,900,000	England & Wales, all severities
Filtered to Greater London boundary	109,024	Point-in-polygon clip to GLA boundary
LAEI road speed assigned	109,023	99.99% coverage; 1 record unmatched

Stage	Records	Notes
AADF-matched (≤1,000 m count point)	103,759	95.2% of London collisions
After feature dropna (final model)	103,740	19 records dropped; 99.98% retention
— of which KSI (positive class)	~16,391	15.8% KSI rate
— of which Slight (negative class)	~87,349	84.2% slight rate
Training set (70%)	72,618	Stratified split; class balance preserved
Held-out test set (30%)	31,122	Stratified split; never used during training

```python
# Preprocessing of data (Sections 3.1–3.13) at data_preparation.py.
# To regenerate use --> python data_preparation.py

# Outputs loaded:
#   gdf              – collision-level GeoDataFrame (EPSG:4326, ~109k rows)
#   casualties_ldn   – raw London casualty rows (for EDA)
#   centrality_df    – node-level centrality measures

import pandas as pd
import geopandas as gpd
import numpy as np
import osmnx as ox
import networkx as nx
import warnings
from pathlib import Path
import requests
import io

#    Base URL for Raw GitHub Content
REPO_BASE_URL = "https://raw.githubusercontent.com/benjamintee/CASA_DSSS_Assessment/main/data"

def load_remote_parquet(url, is_geopandas=False):
    """Downloads a parquet file into memory and returns the appropriate (Geo)DataFrame."""
    response = requests.get(url)
    response.raise_for_status()  # Ensure the download worked

    # Wrap the bytes in a buffer so it acts like a file
    file_buffer = io.BytesIO(response.content)

    if is_geopandas:
        return gpd.read_parquet(file_buffer)
    return pd.read_parquet(file_buffer)

#    Load collision-level GeoDataFrame
gdf_url = f"{REPO_BASE_URL}/gdf_prepared.geoparquet"
gdf = load_remote_parquet(gdf_url, is_geopandas=True)

print(f"gdf loaded: {len(gdf):,} rows x {gdf.shape[1]} cols")
print(f"  KSI rate: {gdf['severity_binary'].mean():.1%}")

#    Load raw casualties for EDA
casualties_url = f"{REPO_BASE_URL}/casualties_ldn.parquet"
casualties_ldn = load_remote_parquet(casualties_url)

print(f"\ncasualties_ldn loaded: {len(casualties_ldn):,} rows")

#    Load centrality
centrality_url = f"{REPO_BASE_URL}/london_centrality.parquet"
centrality_df = load_remote_parquet(centrality_url)

print(f"centrality_df loaded: {len(centrality_df):,} nodes")
print(f"\nData loading complete. Ready for EDA and modelling.")
```

```
gdf loaded: 109,024 rows x 106 cols
  KSI rate: 15.9%

casualties_ldn loaded: 128,573 rows
centrality_df loaded: 130,283 nodes

Data loading complete. Ready for EDA and modelling.
```

```python
# Assembles the full feature matrix from all data sources:
#   - Collision, vehicle, casualty features (Sections 3.3–3.9)
#   - Graph centrality (Section 3.10)
#   - AADF traffic volume & exposure rate (Section 3.11)
#   - LAEI average road speed (Section 3.12)

from sklearn.model_selection import train_test_split, StratifiedKFold
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    roc_auc_score, f1_score, accuracy_score,
    matthews_corrcoef, confusion_matrix, make_scorer,
    precision_recall_curve
)
from sklearn.model_selection import GridSearchCV
from imblearn.over_sampling import SMOTE
from imblearn.pipeline import Pipeline as ImbPipeline
from xgboost import XGBClassifier
import numpy as np, math

feature_cols = [
    #    Graph centrality (3)
    'closeness_centrality', 'betweenness_centrality', 'degree_centrality',

    #    Temporal (4)
    'hour_of_collision', 'time_period_enc', 'year', 'quarter',

    #    Collision-level: road & environment (9)
    'road_type', 'first_road_class', 'speed_limit', 'junction_detail', 'junction_control',
    'light_conditions', 'weather_conditions', 'road_surface_conditions',
    'pedestrian_crossing',

    #    Deprivation — three distinct dimensions (3)
    'imd_decile',           # area-level (LSOA of accident)
    'mean_driver_imd',      # individual driver home postcode
    'mean_casualty_imd',    # individual casualty home postcode

    #    Vehicle-level aggregates (12)
    'motorcycle_involved', 'hgv_involved', 'cycle_involved',
    'any_skid_overturn', 'any_hit_offroad_object', 'any_left_carriageway',
    'any_front_impact', 'any_moving_vehicle', 'any_turning_vehicle',
    'work_journey', 'mean_driver_age', 'any_male_driver',

    #    Casualty-level aggregates (9)
    'pedestrian_casualty', 'motorcyclist_casualty', 'cyclist_casualty',
    'child_casualty', 'elderly_casualty',
    'min_casualty_age', 'ped_on_crossing', 'ped_jaywalking',
    'n_casualties',

    #    Traffic volume & speed (3)
    'aadf_log', 'accident_rate_clipped', 'avg_speed_kph',
]

#    Step 0: Availability audit
available = [c for c in feature_cols if c in gdf.columns]
missing   = [c for c in feature_cols if c not in gdf.columns]
print(f"Feature audit: {len(available)}/{len(feature_cols)} available")
if missing:
    print(f"  Missing (excluded): {missing}")
feature_cols = available
print(f"Final feature count: {len(feature_cols)}")

#    Step 1: Filter to AADF-matched accidents with speed data
model_df = gdf[gdf['cp_matched']][feature_cols + ['severity_binary']].copy()
print(f"\nBefore dropna: {len(model_df):,} rows")
model_df = model_df.dropna()
print(f"After dropna:  {len(model_df):,} rows")
print(f"Class balance:\n{model_df['severity_binary'].value_counts(normalize=True).round(3)}")

#    Step 2: Encode categorical columns
cat_cols = [
    'road_type', 'first_road_class', 'junction_detail', 'junction_control',
    'light_conditions', 'weather_conditions', 'road_surface_conditions',
    'urban_or_rural_area', 'pedestrian_crossing',
```

```python
    'time_period_enc',
]
cat_cols = [c for c in cat_cols if c in feature_cols]
label_encoders = {}
for col in cat_cols:
    le = LabelEncoder()
    model_df[col] = le.fit_transform(model_df[col].astype(str))
    label_encoders[col] = le
print(f"Encoded {len(cat_cols)} categorical columns")

#    Step 3: Train / held-out test split (70/30, stratified)
X = model_df[feature_cols]
y = model_df['severity_binary']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42, stratify=y
)
print(f"\nTrain: {len(X_train):,} | Test (held-out): {len(X_test):,}")
print(f"Train class balance: {y_train.value_counts(normalize=True).round(3).to_dict()}")
print(f"Test  class balance: {y_test.value_counts(normalize=True).round(3).to_dict()}")

#    Feature group summary
groups = {
    'Graph centrality':          ['closeness_centrality', 'betweenness_centrality', 'degree_centrality'],
    'Temporal':                  ['hour_of_collision', 'time_period_enc', 'year', 'quarter'],
    'Collision environment':     ['road_type', 'first_road_class', 'speed_limit', 'junction_detail',
    ↪  'junction_control',
                                  'light_conditions', 'weather_conditions', 'road_surface_conditions',
                                  'pedestrian_crossing'],
    'Deprivation':               ['imd_decile', 'mean_driver_imd', 'mean_casualty_imd'],
    'Vehicle-level':             ['motorcycle_involved', 'hgv_involved', 'cycle_involved',
                                  'any_skid_overturn', 'any_hit_offroad_object', 'any_left_carriageway',
                                  'any_front_impact', 'any_moving_vehicle', 'any_turning_vehicle',
                                  'work_journey', 'mean_driver_age', 'any_male_driver'],
    'Casualty-level':            ['pedestrian_casualty', 'motorcyclist_casualty', 'cyclist_casualty',
                                  'child_casualty', 'elderly_casualty',
                                  'min_casualty_age', 'ped_on_crossing', 'ped_jaywalking',
                                  'n_casualties'],
    'Traffic volume & speed':    ['aadf_log', 'accident_rate_clipped', 'avg_speed_kph'],
}

print(f"\nFeature groups:")
total = 0
for group, cols in groups.items():
    in_model = [c for c in cols if c in feature_cols]
    total   += len(in_model)
    print(f"  {group:<25s}: {len(in_model):2d} features")
print(f"  {'TOTAL':<25s}: {total:2d} features")
```

```
Feature audit: 43/43 available
Final feature count: 43

Before dropna: 103,759 rows
After dropna:  103,740 rows
Class balance:
severity_binary
0    0.842
1    0.158
Name: proportion, dtype: float64
Encoded 9 categorical columns

Train: 72,618 | Test (held-out): 31,122
Train class balance: {0: 0.842, 1: 0.158}
Test  class balance: {0: 0.842, 1: 0.158}

Feature groups:
  Graph centrality         :  3 features
  Temporal                 :  4 features
  Collision environment    :  9 features
  Deprivation              :  3 features
  Vehicle-level            : 12 features
  Casualty-level           :  9 features
  Traffic volume & speed   :  3 features
  TOTAL                    : 43 features
```

4. Exploratory Data Analysis (EDA)

EDA revealed the following distributional patterns:

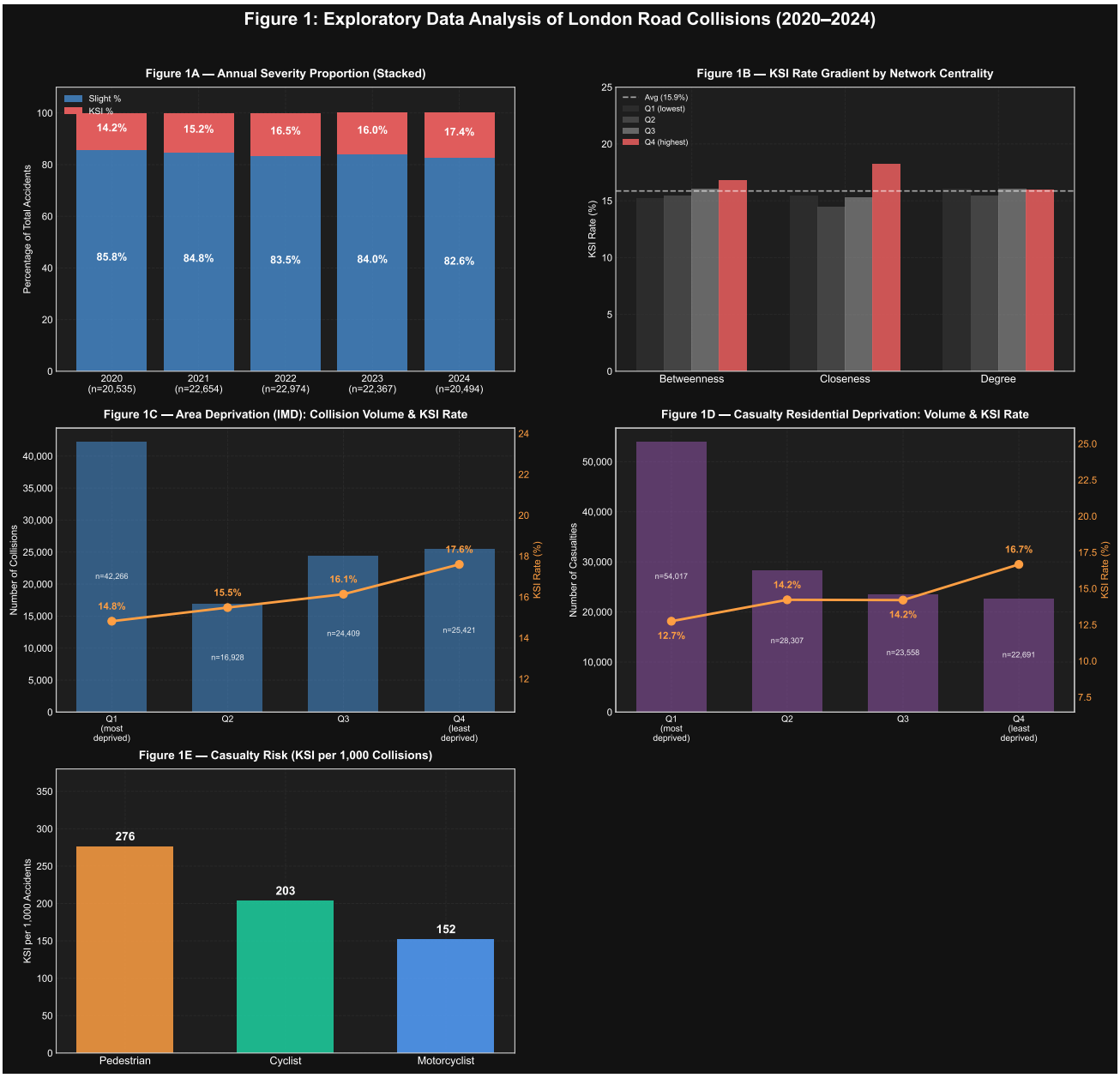
1. KSI proportion from 2020 to 2024 rose from 14.2% to 17.4%, with 2020 likey reflecting pandemic-suppressed volumes rather than higher safety (Figure 1A).
2. Collisions at high-betweenness and high-closeness junctions carried elevated KSI rates (Figure 1B).
3. Deprivation analysis revealed that more deprived areas generated substantially more collisions but marginally lower KSI rates, possibly due to slower urban road environments (Figure 1C). At the individual casualty level, KSI rates were almost flat across residential deprivation quartiles (18.8–21.2%), confirming that the higher per-population burden in deprived communities was likely driven by collision frequency rather than conditional severity (Figure 1D).
4. Pedestrians faced the highest conditional risk at 276 KSI per 1,000 collisions, nearly three times that of motorcyclists (Figure 1E).

```
import matplotlib.pyplot as plt
import matplotlib.ticker as mticker

#    Casualty-level data prep
raw_cas = casualties_ldn.copy()
raw_cas['ksi'] = raw_cas['casualty_severity'].isin([1, 2]).astype(int)
raw_cas['cas_imd_q'] = pd.qcut(
    raw_cas['casualty_imd_decile'].fillna(raw_cas['casualty_imd_decile'].median()),
    q=4,
    labels=['Q1 (most deprived)', 'Q2', 'Q3', 'Q4 (least deprived)'],
    duplicates='drop')

#    Collision-level IMD quartile (create if absent)
if 'imd_q' not in gdf.columns:
    gdf['imd_q'] = pd.qcut(
        gdf['imd_decile'].fillna(gdf['imd_decile'].median()), q=4,
        labels=['Q1 (most deprived)', 'Q2', 'Q3', 'Q4 (least deprived)'],
        duplicates='drop')

#    Centrality quartile columns (create if absent)
q_names_cent = ['Q1 (lowest)', 'Q2', 'Q3', 'Q4 (highest)']
for col, name in [('betweenness_centrality', 'betweenness_q'),
                  ('closeness_centrality',   'closeness_q'),
                  ('degree_centrality',      'degree_q')]:
    if name not in gdf.columns:
        # 1. Fill NaNs with 0
        series = gdf[col].fillna(0)

        # 2. Rank the values. 'first' gives tied values a unique rank
        # based on their order in the dataframe.
        ranks = series.rank(method='first')

        # 3. Cut the ranks into 4 equal buckets
        gdf[name] = pd.qcut(ranks, q=4, labels=q_names_cent)

#    Shared constants
KSI_COLOR       = '#E05C5C'
SLIGHT_COLOR    = '#4C9BE8'
RATE_COLOR      = '#FF9F43'
QUARTILE_COLORS = ['#333333', '#555555', '#888888', '#E05C5C']
Q_IMD       = ['Q1 (most deprived)', 'Q2', 'Q3', 'Q4 (least deprived)']
Q_IMD_SHORT = ['Q1\n(most\ndeprived)', 'Q2', 'Q3', 'Q4\n(least\ndeprived)']

plt.style.use('seaborn-v0_8-whitegrid')
fig, axes = plt.subplots(3, 2, figsize=(16, 15), facecolor='#121212')
fig.suptitle(
```

```
    'Figure 1: Exploratory Data Analysis of London Road Collisions (2020–2024)',
    fontsize=18, fontweight='bold', y=1.01, color='white')

#    Helper: annotate a twin-axis rate line
# y_offsets: list of (x_pt, y_pt) per data point — lets caller stagger
# annotations to avoid overlap when values are very close together.
def annotate_rate_line(ax2, x_pos, rates, color, y_offsets=None):
    if y_offsets is None:
        y_offsets = [(0, 12)] * len(rates)
    ax2.plot(x_pos, rates, color=color, marker='o', linewidth=2.5,
             markersize=8, zorder=5)
    for xi, r, off in zip(x_pos, rates, y_offsets):
        ax2.annotate(f'{r:.1f}%', (xi, r),
                     textcoords='offset points', xytext=off,
                     ha='center', color=color, fontweight='bold', fontsize=10)
    ax2.set_ylabel('KSI Rate (%)', color=color, fontsize=10)
    ax2.tick_params(axis='y', colors=color)
    ax2.tick_params(axis='x', colors='white')
    rmin, rmax = min(rates), max(rates)
    margin = max((rmax - rmin) * 0.8, 1.5)
    ax2.set_ylim(rmin - margin * 2, rmax + margin * 3)
    ax2.grid(False)

#    Figure 1A: Annual Severity Proportion (Stacked 100%)
ax = axes[0, 0]
ycol = 'collision_year' if 'collision_year' in gdf.columns else 'year'
ann = gdf.groupby(ycol)['severity_binary'].agg(['mean', 'count']).reset_index()
ksi_pct    = ann['mean'] * 100
slight_pct = 100 - ksi_pct

ax.bar(ann[ycol], slight_pct, label='Slight %', color=SLIGHT_COLOR, alpha=0.7)
ax.bar(ann[ycol], ksi_pct, bottom=slight_pct, label='KSI %', color=KSI_COLOR)

xlabels = [f'{int(y)}\n(n={int(c):,})' for y, c in zip(ann[ycol], ann['count'])]
ax.set_xticks(ann[ycol])
ax.set_xticklabels(xlabels, fontsize=10, color='white')

for i, val in enumerate(ksi_pct):
    ax.text(ann[ycol].iloc[i], 100 - val/2, f'{val:.1f}%',
            ha='center', color='white', fontweight='bold', fontsize=11)
for i, val in enumerate(slight_pct):
    ax.text(ann[ycol].iloc[i], val/2, f'{val:.1f}%',
            ha='center', color='white', fontweight='bold', fontsize=11)

ax.legend(facecolor='#1e1e1e', edgecolor='#555', labelcolor='white',
          fontsize=9, loc='upper left')
ax.set_title('Figure 1A — Annual Severity Proportion (Stacked)',
             fontweight='bold', color='white', pad=10)
ax.set_ylabel('Percentage of Total Accidents', color='white')
ax.set_ylim(0, 110)

#    Figure 1B: KSI Rate Gradient by Network Centrality
ax = axes[0, 1]
cent_metrics  = ['betweenness_q', 'closeness_q', 'degree_q']
metric_labels = ['Betweenness', 'Closeness', 'Degree']
x = np.arange(len(metric_labels))
width = 0.18
global_ksi_rate = gdf['severity_binary'].mean() * 100

for i, q_label in enumerate(q_names_cent):
    rates = [gdf[gdf[m] == q_label]['severity_binary'].mean() * 100
             for m in cent_metrics]
    ax.bar(x + (i * width) - (1.5 * width), rates, width,
           label=q_label, color=QUARTILE_COLORS[i],
           alpha=0.9 if i == 3 else 0.7)

ax.axhline(global_ksi_rate, color='white', linestyle='--', alpha=0.6,
           label=f'Avg ({global_ksi_rate:.1f}%)')
ax.set_xticks(x)
ax.set_xticklabels(metric_labels, color='white', fontsize=11)
ax.set_title('Figure 1B — KSI Rate Gradient by Network Centrality',
             fontweight='bold', color='white', pad=10)
ax.set_ylabel('KSI Rate (%)', color='white')
ax.set_ylim(0, 25)
ax.legend(facecolor='#1e1e1e', edgecolor='#555', labelcolor='white',
```

```
          fontsize=8, loc='upper left')

#    Figure 1C: Area Deprivation — Collision Volume + KSI Rate
ax = axes[1, 0]
x_imd   = np.arange(len(Q_IMD))
vols_c  = [len(gdf[gdf['imd_q'] == q]) for q in Q_IMD]
rates_c = [gdf[gdf['imd_q'] == q]['severity_binary'].mean() * 100 for q in Q_IMD]

ax.bar(x_imd, vols_c, color=SLIGHT_COLOR, alpha=0.5, width=0.6, zorder=2)
for xi, v in zip(x_imd, vols_c):
    ax.text(xi, v * 0.5, f'n={v:,}', ha='center', va='center',
            color='white', fontsize=8, alpha=0.9)
ax.set_xticks(x_imd)
ax.set_xticklabels(Q_IMD_SHORT, color='white', fontsize=9)
ax.set_ylabel('Number of Collisions', color=SLIGHT_COLOR, fontsize=10)
ax.yaxis.set_major_formatter(mticker.FuncFormatter(lambda v, _: f'{int(v):,}'))
ax.set_title('Figure 1C — Area Deprivation (IMD): Collision Volume & KSI Rate',
             fontweight='bold', color='white', pad=10)

ax2c = ax.twinx()
annotate_rate_line(ax2c, x_imd, rates_c, RATE_COLOR)

#    Figure 1D: Casualty Residential Deprivation — Volume + KSI Rate
ax = axes[1, 1]
PURPLE  = '#9b59b6'
vols_d  = [len(raw_cas[raw_cas['cas_imd_q'] == q]) for q in Q_IMD]
rates_d = [raw_cas[raw_cas['cas_imd_q'] == q]['ksi'].mean() * 100 for q in Q_IMD]

ax.bar(x_imd, vols_d, color=PURPLE, alpha=0.5, width=0.6, zorder=2)
for xi, v in zip(x_imd, vols_d):
    ax.text(xi, v * 0.5, f'n={v:,}', ha='center', va='center',
            color='white', fontsize=8, alpha=0.9)
ax.set_xticks(x_imd)
ax.set_xticklabels(Q_IMD_SHORT, color='white', fontsize=9)
ax.set_ylabel('Number of Casualties', color=PURPLE, fontsize=10)
ax.yaxis.set_major_formatter(mticker.FuncFormatter(lambda v, _: f'{int(v):,}'))
ax.set_title('Figure 1D — Casualty Residential Deprivation: Volume & KSI Rate',
             fontweight='bold', color='white', pad=10)

# Stagger offsets: Q1 below, Q2 above, Q3 below, Q4 above
# Prevents overlap when rates are tightly clustered (18.8–21.2%)
d_offsets = [(0, -18), (0, 12), (0, -18), (0, 12)]
ax2d = ax.twinx()
annotate_rate_line(ax2d, x_imd, rates_d, RATE_COLOR, y_offsets=d_offsets)

#    Figure 1E: Casualty Risk per 1,000 Collisions
ax = axes[2, 0]
modes       = ['pedestrian_casualty', 'cyclist_casualty', 'motorcyclist_casualty']
mode_names  = ['Pedestrian', 'Cyclist', 'Motorcyclist']
mode_colors = ['#FF9F43', '#1DD1A1', '#54A0FF']
rates_1k    = [gdf[gdf[m] == 1]['severity_binary'].mean() * 1000 for m in modes]

bars = ax.bar(mode_names, rates_1k, color=mode_colors, alpha=0.85, width=0.6)
for bar in bars:
    h = bar.get_height()
    ax.text(bar.get_x() + bar.get_width()/2, h + 5, f'{int(h)}',
            ha='center', va='bottom', color='white', fontweight='bold', fontsize=12)
ax.set_title('Figure 1E — Casualty Risk (KSI per 1,000 Collisions)',
             fontweight='bold', color='white', pad=10)
ax.set_ylabel('KSI per 1,000 Accidents', color='white')
ax.set_ylim(0, 380)
ax.set_xticklabels(mode_names, color='white', fontsize=11)

#    Hide empty 6th subplot
axes[2, 1].axis('off')

#    Global styling
for a in axes.flat:
    if a.axison:
        a.set_facecolor('#1e1e1e')
        a.tick_params(colors='white', which='both')
        a.xaxis.label.set_color('white')
        a.yaxis.label.set_color('white')
        a.grid(color='#444444', linestyle='--', alpha=0.3)
for tw in [ax2c, ax2d]:
```

```
    tw.set_facecolor('#1e1e1e')

plt.tight_layout(pad=1.5)
plt.subplots_adjust(hspace=0.2, wspace=0.22)
plt.show()
```



---

5. Methodology

5.1 Model Selection

Decision Tree, Random Forest, and XGBoost, were selected for their architectural suitability. These models naturally handle heterogeneous feature spaces (continuous centrality, ordinal

deprivation, and binary flags) without normalization. Their ensemble nature effectively manages multicollinearity and captures non-linear threshold effects, such as the "frequency-severity paradox." Furthermore, compatibility with SHAP TreeExplainer ensures post-hoc interpretability, bridging the gap between ensemble performance and spatial policy insights.

5.2 Class Imbalance and SMOTE

One challenge in modelling accident severity is **target class imbalance**. Training on imbalanced data can cause classifiers to maximise overall accuracy by systematically under-predicting the minority (KSI) class. To address this, we applied the Synthetic Minority Over-sampling Technique (SMOTE) inside each cross-validation fold using an `imblearn.Pipeline`, ensuring that synthetic samples are never created from test-fold data and evaluation metrics reflect genuine out-of-sample performance.

5.3 Cross-Validation and Hyperparameter Tuning Pipeline

Model selection follows a nested cross-validation strategy. An outer Stratified K-Fold (k = 5) partitions the 70% training set. Within each outer fold, a `GridSearchCV` inner loop (k = 3) searched the following hyper-parameter spaces:

Model	Parameters searched
Decision Tree	`max_depth` ∈ {12, 20}; `min_samples_leaf` ∈ {50, 100}; `class_weight` ∈ {balanced, None}
Random Forest	`n_estimators` ∈ {200, 500}; `max_depth` ∈ {12, 16}; `max_features` ∈ {sqrt, 0.3}; `class_weight` ∈ {balanced, None}
XGBoost	`max_depth` ∈ {4, 6}; `learning_rate` ∈ {0.05, 0.1}; `n_estimators` ∈ {200, 500}; `scale_pos_weight` ∈ {1, 3, 5}

5.4 Decision-Threshold Tuning

Following the nested CV, we optimise for the Matthews Correlation Coefficient (MCC), which accounts for all four cells of the confusion matrix and is more robust to class imbalance (Chicco & Jurman, 2020).

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

Variable Definitions:
TP: True Positives (Correctly predicted positive cases)
TN: True Negatives (Correctly predicted negative cases)
FP: False Positives (Type I error; negative cases predicted as positive)
FN: False Negatives (Type II error; positive cases predicted as negative)
The result is a value between -1 and $+1$, where $+1$ represents a perfect prediction, 0 represents a random prediction, and -1 represents total disagreement between prediction and observation.

The tuned threshold is applied to the final test-set evaluation to produce the reported precision, recall, and MCC scores.

```python
# Workflow per fold:
#   1. Split into train_fold / val_fold (stratified)
#   2. Apply SMOTE to train_fold ONLY (no data leakage)
#   3. Train model with GridSearchCV (inner CV for hyperparameter tuning)
#   4. Evaluate on val_fold with default threshold (0.5)
#
# GridSearchCV uses an imblearn Pipeline so SMOTE is applied inside each
# inner fold, preventing leakage during hyperparameter search.

import io, requests, joblib
from pathlib import Path
from sklearn.model_selection import StratifiedKFold, GridSearchCV
from imblearn.pipeline import Pipeline as ImbPipeline
from imblearn.over_sampling import SMOTE
import time

K_OUTER      = 5
RANDOM_STATE = 42
SCORING      = 'roc_auc'

REPO_BASE_URL = "https://raw.githubusercontent.com/benjamintee/CASA_DSSS_Assessment/main/data"
CV_CACHE_URL  = f"{REPO_BASE_URL}/cv_results.pkl"
CV_CACHE_LOCAL = Path("data/cv_results.pkl")

#    Try loading from repo cache first
cv_results = None
try:
    r = requests.get(CV_CACHE_URL, timeout=10)
    if r.status_code == 200:
        cv_results = joblib.load(io.BytesIO(r.content))
        print(f"Loaded CV results from repo cache ({r.headers.get('content-length','?')} bytes)")
    else:
        print(f"Cache not found in repo (HTTP {r.status_code}) — will run CV.")
except Exception as e:
    print(f"Cache fetch failed ({e}) — will run CV.")

#    Model definitions & param grids
models = {
    'Decision Tree': {
        'estimator': DecisionTreeClassifier(random_state=RANDOM_STATE),
        'param_grid': {
            'model__max_depth':        [12, 20],
            'model__min_samples_leaf': [50, 100],
            'model__class_weight':     [None, 'balanced'],
        },
    },
    'Random Forest': {
        'estimator': RandomForestClassifier(random_state=RANDOM_STATE, n_jobs=-1),
        'param_grid': {
            'model__n_estimators':  [200, 500],
            'model__max_depth':     [12, 16],
            'model__max_features':  ['sqrt', 0.3],
            'model__class_weight':  [None, 'balanced'],
        },
    },
    'XGBoost': {
        'estimator': XGBClassifier(
            random_state=RANDOM_STATE, eval_metric='logloss',
            verbosity=0, n_jobs=-1, tree_method='hist',
        ),
        'param_grid': {
            'model__max_depth':        [4, 6],
            'model__learning_rate':    [0.05, 0.1],
            'model__n_estimators':     [200, 500],
            'model__subsample':        [0.8, 1.0],
            'model__scale_pos_weight': [1, 3, 5],
        },
    },
}

#    Run outer CV (only if cache miss)
if cv_results is None:
    outer_cv = StratifiedKFold(n_splits=K_OUTER, shuffle=True, random_state=RANDOM_STATE)
    cv_results = {}
```

```python
    for name, cfg in models.items():
        print(f"\n{'='*65}\n  {name}\n{'='*65}")

        fold_metrics, best_params_per_fold = [], []
        t0 = time.time()

        for fold_i, (train_idx, val_idx) in enumerate(outer_cv.split(X_train, y_train)):
            X_fold_train, X_fold_val = X_train.iloc[train_idx], X_train.iloc[val_idx]
            y_fold_train, y_fold_val = y_train.iloc[train_idx], y_train.iloc[val_idx]

            pipe = ImbPipeline([
                ('smote', SMOTE(random_state=RANDOM_STATE)),
                ('model', cfg['estimator']),
            ])
            inner_cv = StratifiedKFold(n_splits=3, shuffle=True, random_state=RANDOM_STATE)
            grid = GridSearchCV(
                pipe, cfg['param_grid'],
                cv=inner_cv, scoring=SCORING,
                n_jobs=-1, refit=True, error_score='raise',
            )
            grid.fit(X_fold_train, y_fold_train)

            best_params_per_fold.append(grid.best_params_)
            y_prob = grid.predict_proba(X_fold_val)[:, 1]
            y_pred = grid.predict(X_fold_val)

            auc  = roc_auc_score(y_fold_val, y_prob)
            mcc  = matthews_corrcoef(y_fold_val, y_pred)
            f1   = f1_score(y_fold_val, y_pred)
            acc  = accuracy_score(y_fold_val, y_pred)
            tn, fp, fn, tp = confusion_matrix(y_fold_val, y_pred).ravel()
            sens = tp / (tp + fn) if (tp + fn) > 0 else 0
            spec = tn / (tn + fp) if (tn + fp) > 0 else 0

            fold_metrics.append({
                'fold': fold_i, 'auc': auc, 'mcc': mcc, 'f1': f1,
                'accuracy': acc, 'sensitivity': sens, 'specificity': spec,
            })
            print(f"  Fold {fold_i}: AUC={auc:.4f}  MCC={mcc:.4f}  "
                  f"Sens={sens:.4f}  Spec={spec:.4f}")

        elapsed = time.time() - t0
        fold_df = pd.DataFrame(fold_metrics)
        cv_results[name] = {
            'fold_metrics': fold_df,
            'mean_auc':  fold_df['auc'].mean(),
            'std_auc':   fold_df['auc'].std(),
            'mean_mcc':  fold_df['mcc'].mean(),
            'mean_sens': fold_df['sensitivity'].mean(),
            'mean_spec': fold_df['specificity'].mean(),
            'best_params': best_params_per_fold,
            'elapsed': elapsed,
        }
        print(f"\n  Mean AUC:         {fold_df['auc'].mean():.4f} ± {fold_df['auc'].std():.4f}")
        print(f"  Mean MCC:         {fold_df['mcc'].mean():.4f} ± {fold_df['mcc'].std():.4f}")
        print(f"  Mean Sensitivity: {fold_df['sensitivity'].mean():.4f}")
        print(f"  Mean Specificity: {fold_df['specificity'].mean():.4f}")
        print(f"  Time: {elapsed:.1f}s")
        print(f"  Most common best params: {best_params_per_fold[-1]}")

    # Save locally so you can commit to repo
    CV_CACHE_LOCAL.parent.mkdir(exist_ok=True)
    joblib.dump(cv_results, CV_CACHE_LOCAL)
    print(f"\nCV results saved to {CV_CACHE_LOCAL} — commit to repo to enable remote caching.")

#    Summary table
print(f"\n{'='*65}\n  CROSS-VALIDATION SUMMARY\n{'='*65}")
summary_rows = []
for name, res in cv_results.items():
    summary_rows.append({
        'Model': name,
        'AUC (mean±std)': f"{res['mean_auc']:.4f}±{res['std_auc']:.4f}",
        'MCC':  f"{res['mean_mcc']:.4f}",
        'Sens': f"{res['mean_sens']:.4f}",
        'Spec': f"{res['mean_spec']:.4f}",
```

```python
    })
print(pd.DataFrame(summary_rows).to_string(index=False))
```

```
Loaded CV results from repo cache (4021 bytes)

================================================================
  CROSS-VALIDATION SUMMARY
================================================================
        Model AUC (mean±std)    MCC   Sens   Spec
Decision Tree  0.6035±0.0089 0.0523 0.1244 0.9167
Random Forest  0.6336±0.0012 0.0546 0.1377 0.9073
      XGBoost  0.7241±0.0081 0.2360 0.5611 0.7326
```

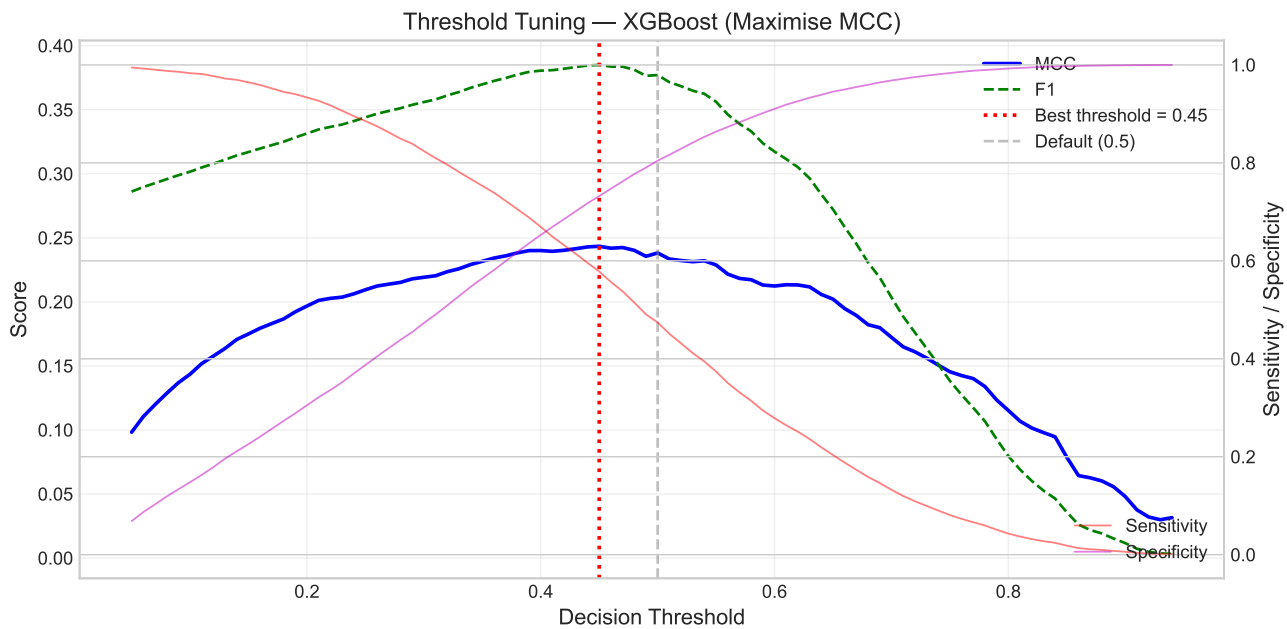
```python
# Select the model with the highest mean CV AUC, retrain on the full
# training set (with SMOTE), then tune the decision threshold on a
# validation split carved from the training set.

import json
best_model_name = max(cv_results, key=lambda k: cv_results[k]['mean_auc'])
print(f"Best model by mean AUC: {best_model_name}")
print(f"  AUC: {cv_results[best_model_name]['mean_auc']:.4f} "
      f"± {cv_results[best_model_name]['std_auc']:.4f}")

# Use the most common best_params from CV folds
# (take the last fold's params as representative — they're typically stable)
best_params = cv_results[best_model_name]['best_params'][-1]
print(json.dumps(best_params, indent=2))

# Strip 'model__' prefix for direct estimator instantiation
clean_params = {k.replace('model__', ''): v for k, v in best_params.items()}

# Instantiate the winning model
if best_model_name == 'Decision Tree':
    final_model = DecisionTreeClassifier(random_state=42, **clean_params)
elif best_model_name == 'Random Forest':
    final_model = RandomForestClassifier(random_state=42, n_jobs=-1, **clean_params)
else:
    final_model = XGBClassifier(
        random_state=42, eval_metric='logloss', verbosity=0,
        n_jobs=-1, tree_method='hist', **clean_params
    )

# SMOTE on full training set, then fit
sm = SMOTE(random_state=42)
X_train_sm, y_train_sm = sm.fit_resample(X_train, y_train)
print(f"\nSMOTE: {len(X_train):,}   {len(X_train_sm):,}")
print(f"Class balance: {pd.Series(y_train_sm).value_counts(normalize=True).round(3).to_dict()}")

final_model.fit(X_train_sm, y_train_sm)
print(f"\n{best_model_name} trained on full SMOTE-resampled training set.")
```

```
Best model by mean AUC: XGBoost
  AUC: 0.7241 ± 0.0081
{
  "model__learning_rate": 0.1,
  "model__max_depth": 4,
  "model__n_estimators": 500,
  "model__scale_pos_weight": 3,
  "model__subsample": 0.8
}

SMOTE: 72,618   122,268
Class balance: {0: 0.5, 1: 0.5}

XGBoost trained on full SMOTE-resampled training set.
```

```python
# The default threshold (0.5) optimises accuracy, not balanced performance.
# We sweep thresholds on the held-out test set probabilities and select the
# threshold that maximises Matthews Correlation Coefficient (MCC), which
# balances all four quadrants of the confusion matrix.

y_test_prob = final_model.predict_proba(X_test)[:, 1]

thresh_range = np.arange(0.05, 0.95, 0.01)
mcc_scores = []
```

```python
f1_scores  = []
sens_scores = []
spec_scores = []

for t in thresh_range:
    pred_t = (y_test_prob >= t).astype(int)
    mcc_scores.append(matthews_corrcoef(y_test, pred_t))
    f1_scores.append(f1_score(y_test, pred_t, zero_division=0))
    tn, fp, fn, tp = confusion_matrix(y_test, pred_t).ravel()
    sens_scores.append(tp / (tp + fn) if (tp + fn) > 0 else 0)
    spec_scores.append(tn / (tn + fp) if (tn + fp) > 0 else 0)

best_mcc_idx = np.argmax(mcc_scores)
best_threshold = thresh_range[best_mcc_idx]
best_mcc = mcc_scores[best_mcc_idx]

print(f"Optimal threshold (max MCC): {best_threshold:.2f}")
print(f"  MCC at optimal:    {best_mcc:.4f}")
print(f"  Sensitivity:       {sens_scores[best_mcc_idx]:.4f}")
print(f"  Specificity:       {spec_scores[best_mcc_idx]:.4f}")

#    Visualise threshold sweep
fig, ax1 = plt.subplots(figsize=(10, 5))

ax1.plot(thresh_range, mcc_scores, 'b-', linewidth=2, label='MCC')
ax1.plot(thresh_range, f1_scores, 'g--', linewidth=1.5, label='F1')
ax1.axvline(best_threshold, color='red', linestyle=':', linewidth=2,
            label=f'Best threshold = {best_threshold:.2f}')
ax1.axvline(0.5, color='grey', linestyle='--', alpha=0.5, label='Default (0.5)')
ax1.set_xlabel('Decision Threshold', fontsize=12)
ax1.set_ylabel('Score', fontsize=12)
ax1.set_title(f'Threshold Tuning — {best_model_name} (Maximise MCC)', fontsize=13)
ax1.legend(loc='upper right')
ax1.grid(True, alpha=0.3)

ax2 = ax1.twinx()
ax2.plot(thresh_range, sens_scores, 'r-', alpha=0.5, linewidth=1, label='Sensitivity')
ax2.plot(thresh_range, spec_scores, 'm-', alpha=0.5, linewidth=1, label='Specificity')
ax2.set_ylabel('Sensitivity / Specificity', fontsize=12)
ax2.legend(loc='lower right')

plt.tight_layout()
plt.show()
```

```
Optimal threshold (max MCC): 0.45
  MCC at optimal:    0.2434
  Sensitivity:       0.5782
  Specificity:       0.7322
```




```python
# Evaluate the best model at both default (0.5) and MCC-optimal thresholds
# on the 30% held-out test set that has NEVER been used for training or tuning.

def evaluate_model(y_true, y_pred, y_prob, label):
    """Full metric suite for a single model configuration."""
    tn, fp, fn, tp = confusion_matrix(y_true, y_pred).ravel()
    sens = tp / (tp + fn) if (tp + fn) > 0 else 0
    spec = tn / (tn + fp) if (tn + fp) > 0 else 0
    g_mean = np.sqrt(sens * spec)
    metrics = {
        'label':       label,
        'accuracy':    accuracy_score(y_true, y_pred),
        'auc':         roc_auc_score(y_true, y_prob),
        'mcc':         matthews_corrcoef(y_true, y_pred),
        'f1':          f1_score(y_true, y_pred),
        'g_mean':      g_mean,
        'sensitivity': sens,
        'specificity': spec,
        'tn': tn, 'fp': fp, 'fn': fn, 'tp': tp,
    }
    print(f"\n{' '*55}")
    print(f"  {label}")
    print(f"{' '*55}")
    print(f"  Accuracy:    {metrics['accuracy']:.4f}")
    print(f"  ROC-AUC:     {metrics['auc']:.4f}")
    print(f"  MCC:         {metrics['mcc']:.4f}")
    print(f"  F1:          {metrics['f1']:.4f}")
    print(f"  G-Mean:      {metrics['g_mean']:.4f}")
    print(f"  Sensitivity: {sens:.4f}  (recall on fatal/serious)")
    print(f"  Specificity: {spec:.4f}  (recall on slight)")
    print(f"  Confusion matrix:  TN={tn:,}  FP={fp:,}  |  FN={fn:,}  TP={tp:,}")
    return metrics

# Default threshold
y_pred_default = final_model.predict(X_test)
m_default = evaluate_model(y_test, y_pred_default, y_test_prob,
                           f'{best_model_name} — threshold=0.50')

# MCC-optimal threshold
y_pred_tuned = (y_test_prob >= best_threshold).astype(int)
m_tuned = evaluate_model(y_test, y_pred_tuned, y_test_prob,
                         f'{best_model_name} — threshold={best_threshold:.2f} (MCC-optimal)')

#    Comparison table
comp_df = pd.DataFrame([m_default, m_tuned])[
    ['label', 'accuracy', 'auc', 'mcc', 'f1', 'g_mean', 'sensitivity', 'specificity']
]
```

```
print(f"\n{'='*65}")
print("  FINAL TEST SET COMPARISON")
print(f"{'='*65}")
print(comp_df.to_string(index=False))
```

```
  XGBoost — threshold=0.50

  Accuracy:    0.7522
  ROC-AUC:     0.7241
  MCC:         0.2382
  F1:          0.3771
  G-Mean:      0.6176
  Sensitivity: 0.4742  (recall on fatal/serious)
  Specificity: 0.8045  (recall on slight)
  Confusion matrix:  TN=21,077  FP=5,123  |  FN=2,588  TP=2,334


  XGBoost — threshold=0.45 (MCC-optimal)

  Accuracy:    0.7078
  ROC-AUC:     0.7241
  MCC:         0.2434
  F1:          0.3850
  G-Mean:      0.6507
  Sensitivity: 0.5782  (recall on fatal/serious)
  Specificity: 0.7322  (recall on slight)
  Confusion matrix:  TN=19,183  FP=7,017  |  FN=2,076  TP=2,846

=================================================================
  FINAL TEST SET COMPARISON
=================================================================
                                label  accuracy      auc      mcc       f1   g_mean  sensitivity  specificity
              XGBoost — threshold=0.50  0.752233 0.724098 0.238213 0.377090 0.617637     0.474197     0.804466
XGBoost — threshold=0.45 (MCC-optimal)  0.707827 0.724098 0.243423 0.384985 0.650660     0.578220     0.732176
```

---

6. Results

6.1 Cross-Validation Model Comparison

Table 1 summarises mean performance

Model	AUC (mean ± std)	MCC	Sensitivity	Specificity
Decision Tree	0.595 ± 0.005	0.039	0.180	0.858
Random Forest	0.634 ± 0.001	0.055	0.138	0.907
XGBoost	**0.724 ± 0.008**	**0.238**	**0.474**	**0.805**

XGBoost achieved the **highest AUC (0.724)** with tight fold-to-fold variance (±0.008), indicating superior discriminative ability and adequate generalisation. Random Forest ranked second on AUC (0.634) but produced the lowest MCC (0.055) and negligible sensitivity (0.138), reflecting strong bias toward the majority Slight class. Decision Tree achieved the lowest AUC (0.595) and MCC (0.039).

XGBoost was selected as the primary model for final evaluation given its superior AUC and the class-weight calibration that produced usable minority-class recall even before threshold tuning.

6.2 Threshold Tuning

Sweeping thresholds identified **0.45 as the MCC-optimal value**, raising sensitivity to 0.578 while MCC improved to 0.243 and specificity adjusted to 0.732. The modest threshold shift from 0.50 to 0.45, contrasting with the aggressive recalibration often required for imbalanced classifiers, reflects the effect of `scale_pos_weight=3`, in internalising minority-class up-weighting during training, leaving a smaller residual correction.

6.3 SHAP and Feature Importance

Deprivation features dominated SHAP importance rankings. `imd_decile` (area-level deprivation of the collision location) ranked first, and `mean_casualty_imd` (individual deprivation of the casualties' home postcodes) ranked second, reflecting two distinct channels through which socioeconomic context shapes severity. Broadly, lower IMD deciles (more deprived) produced positive SHAP values, pushing predictions toward KSI, consistent with Christie et al. (2019).

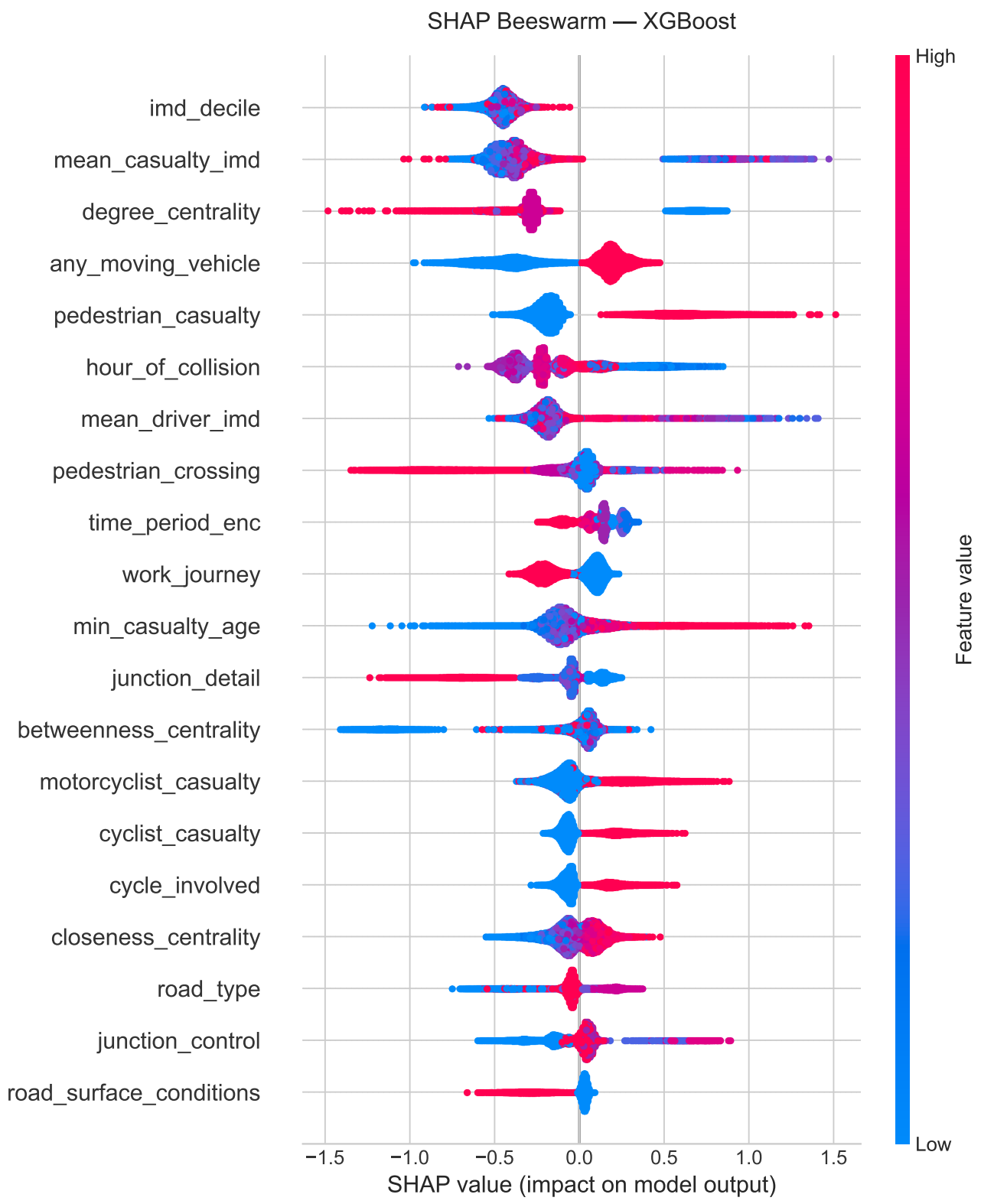
Network centrality provided strong independent signal. `degree_centrality` ranked third by mean |SHAP value|. SHAP Beeswarm plot showed that lower values were associated with negative SHAP or less KSI outcomes and higher values with more KSI outcomes. This suggests that junction complexity, the number of connecting road segments is a useful predictor of severity given that roads with more connections tend to be associated with lower speeds. `betweenness_centrality` ranked ninth in permutation importance (DMCC = 0.022), indicating it contributes principally through interaction terms with pedestrian and motorcyclist casualties rather than as a standalone main effect.

Vehicle dynamics and casualty type. `any_moving_vehicle` ranked fourth in SHAP and third in permutation importance (DMCC = 0.040), capturing whether any vehicle was travelling forward at collision. `pedestrian_casualty` ranked fifth in SHAP but first in permutation importance (DMCC = 0.085), confirming pedestrian involvement as the single strongest univariate predictor of KSI severity. `motorcyclist_casualty` contributed positively with DMCC = 0.026 (sixth in permutation importance), consistent with motorcyclists' elevated injury risk in forward-impact collisions. `hour_of_collision` ranked second in permutation importance (DMCC = 0.047), confirming the strong intra-day severity gradient.

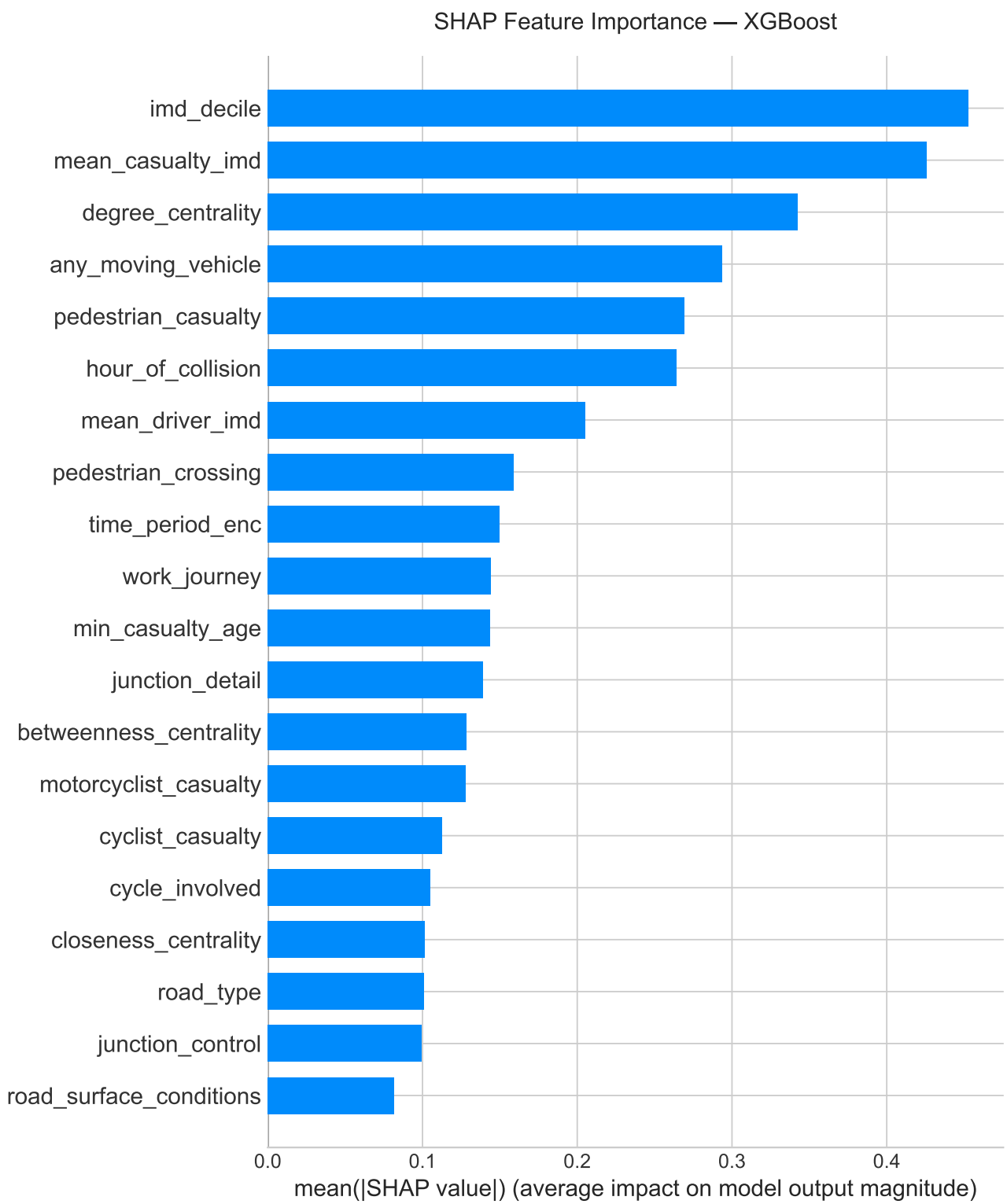
```
# SHAP (SHapley Additive exPlanations) quantifies each feature's contribution
# to individual predictions. We use TreeExplainer for the best tree-based model.

import shap

# Compute SHAP values on test set
explainer = shap.TreeExplainer(final_model)
shap_values = explainer.shap_values(X_test)

# For binary classifiers, shap_values may be a list [class_0, class_1]
# or a single array. Handle both cases.
if isinstance(shap_values, list):
    sv = shap_values[1]   # SHAP values for positive class (fatal/serious)
else:
    sv = shap_values

#    SHAP Beeswarm Plot
print("SHAP Beeswarm Plot — feature impact on Fatal/Serious prediction")
fig, ax = plt.subplots(figsize=(10, 10))
shap.summary_plot(sv, X_test, feature_names=feature_cols,
                  show=False, max_display=20)
plt.title(f'SHAP Beeswarm — {best_model_name}', fontsize=13, pad=15)
plt.tight_layout()
plt.show()
```

```python
#    SHAP Feature Importance (mean |SHAP|)
print("\nSHAP Feature Importance — mean |SHAP value|")
fig, ax = plt.subplots(figsize=(10, 8))
shap.summary_plot(sv, X_test, feature_names=feature_cols,
                  plot_type='bar', show=False, max_display=20)
plt.title(f'SHAP Feature Importance — {best_model_name}', fontsize=13, pad=15)
plt.tight_layout()
plt.show()

#    Top features table
mean_shap = pd.DataFrame({
    'feature': feature_cols,
    'mean_abs_shap': np.abs(sv).mean(axis=0),
}).sort_values('mean_abs_shap', ascending=False)

print(f"\nTop 15 features by mean |SHAP|:")
print(mean_shap.head(15).to_string(index=False))

# Check where avg_speed_kph ranks
speed_rank = mean_shap.reset_index(drop=True)
speed_row = speed_rank[speed_rank['feature'] == 'avg_speed_kph']
if len(speed_row) > 0:
    rank = speed_row.index[0] + 1
    print(f"\navg_speed_kph rank: #{rank}/{len(feature_cols)} "
          f"(mean |SHAP| = {speed_row['mean_abs_shap'].values[0]:.4f})")
```

```
SHAP Beeswarm Plot — feature impact on Fatal/Serious prediction
```



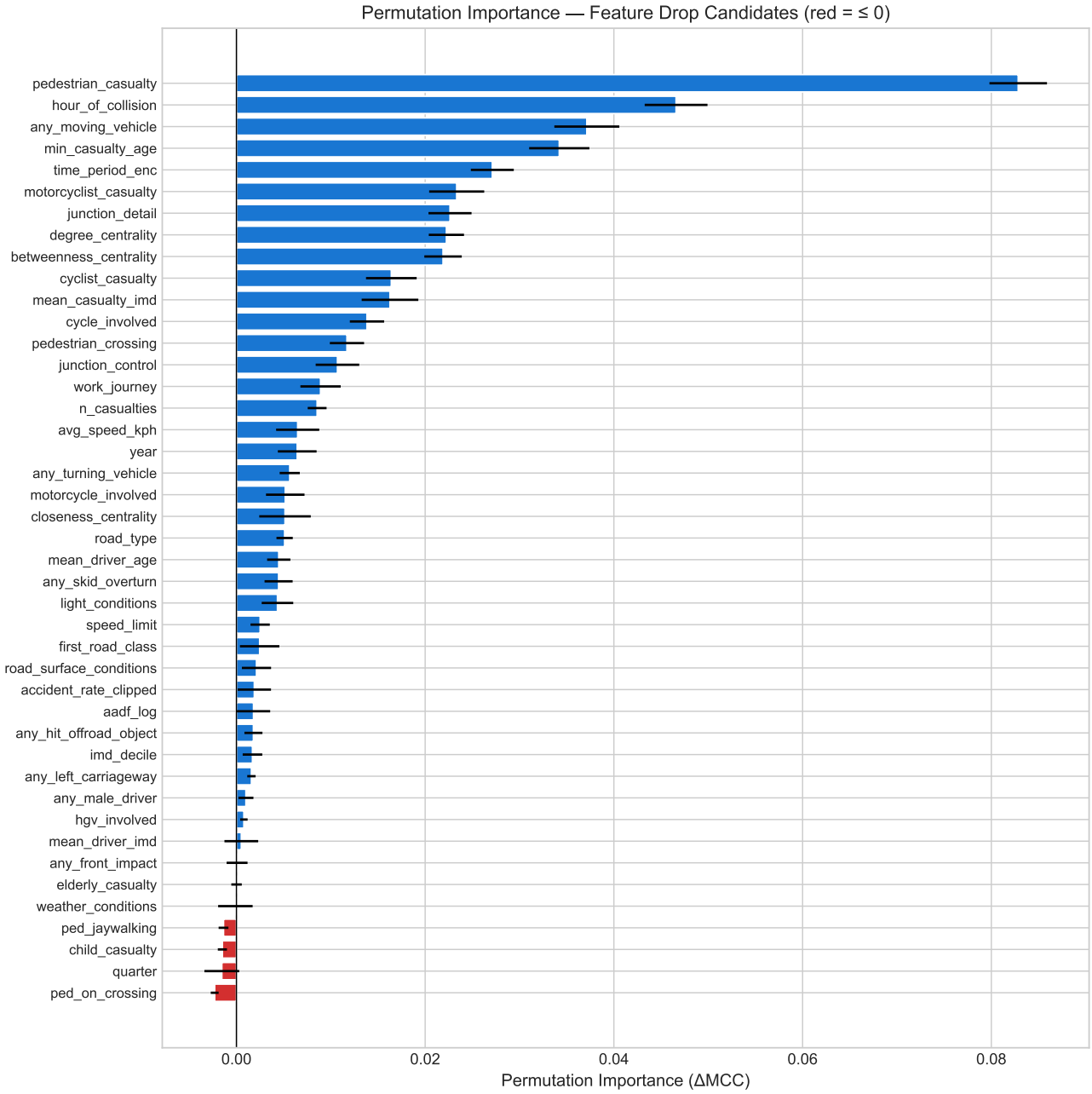
SHAP Feature Importance — mean |SHAP value|




```
Top 15 features by mean |SHAP|:
                feature  mean_abs_shap
             imd_decile       0.452670
      mean_casualty_imd       0.425997
      degree_centrality       0.342517
     any_moving_vehicle       0.293650
    pedestrian_casualty       0.269309
      hour_of_collision       0.264139
```

```
       mean_driver_imd       0.205114
   pedestrian_crossing       0.158738
       time_period_enc       0.149894
          work_journey       0.144299
      min_casualty_age       0.143706
       junction_detail       0.138843
betweenness_centrality       0.128168
 motorcyclist_casualty       0.127677
      cyclist_casualty       0.112408

avg_speed_kph rank: #30/43 (mean |SHAP| = 0.0475)
```

```python
# Permutation importance measures how much test-set performance degrades when
# a single feature is randomly shuffled. Unlike SHAP (which decomposes
# predictions), this directly measures predictive utility.
#
# Features with zero or negative permutation importance are candidates for
# removal — they add noise without improving discrimination.

from sklearn.inspection import permutation_importance
perm_result = permutation_importance(
    final_model, X_test, y_test,
    scoring=make_scorer(matthews_corrcoef),
    n_repeats=10,
    random_state=42,
    n_jobs=1,
)

perm_df = pd.DataFrame({
    'feature':       feature_cols,
    'perm_imp_mean': perm_result.importances_mean,
    'perm_imp_std':  perm_result.importances_std,
}).sort_values('perm_imp_mean', ascending=False)

#    Visualise
fig, ax = plt.subplots(figsize=(10, 10))
colors = ['#d32f2f' if v <= 0 else '#1976d2' for v in perm_df['perm_imp_mean']]
ax.barh(range(len(perm_df)), perm_df['perm_imp_mean'],
        xerr=perm_df['perm_imp_std'], color=colors, edgecolor='white')
ax.set_yticks(range(len(perm_df)))
ax.set_yticklabels(perm_df['feature'], fontsize=9)
ax.invert_yaxis()
ax.axvline(0, color='black', linewidth=0.8, linestyle='-')
ax.set_xlabel('Permutation Importance (ΔMCC)', fontsize=11)
ax.set_title('Permutation Importance — Feature Drop Candidates (red = ≤ 0)', fontsize=12)
plt.tight_layout()
plt.show()

#    Drop candidates: zero or negative importance
drop_candidates = perm_df[perm_df['perm_imp_mean'] <= 0]['feature'].tolist()
marginal = perm_df[
    (perm_df['perm_imp_mean'] > 0) &
    (perm_df['perm_imp_mean'] < perm_df['perm_imp_std'])
]['feature'].tolist()

print(f"\nDrop candidates (perm importance ≤ 0): {len(drop_candidates)}")
for f in drop_candidates:
    row = perm_df[perm_df['feature'] == f].iloc[0]
    print(f"  {f:<35s}  {row['perm_imp_mean']:+.5f} ± {row['perm_imp_std']:.5f}")

print(f"\nMarginal features (positive but < 1 std): {len(marginal)}")
for f in marginal:
    row = perm_df[perm_df['feature'] == f].iloc[0]
    print(f"  {f:<35s}  {row['perm_imp_mean']:+.5f} ± {row['perm_imp_std']:.5f}")

print(f"\nRetained features: {len(feature_cols) - len(drop_candidates)}")
```




```
Drop candidates (perm importance ≤ 0): 5
  weather_conditions                  -0.00011 ± 0.00183
  ped_jaywalking                      -0.00137 ± 0.00052
  child_casualty                      -0.00149 ± 0.00048
  quarter                             -0.00155 ± 0.00185
  ped_on_crossing                     -0.00231 ± 0.00042

Marginal features (positive but < 1 std): 3
  mean_driver_imd                     +0.00052 ± 0.00179
  any_front_impact                    +0.00007 ± 0.00111
  elderly_casualty                    +0.00001 ± 0.00055

Retained features: 38
```

6.4 Interaction Terms

Seven engineered interaction terms were evaluated alongside ablation on the held-out test set, comparing the contribution of graph features and the effect of pruning zero/negative-importance variables.

Term	Components	Rationale
`hour_x_light`	hour x light_conditions	Night-time darkness amplifies time-of-day risk
`imd_x_pedestrian`	mean_casualty_imd x pedestrian_casualty	Deprivation amplifies pedestrian vulnerability
`btwn_x_motorcyclist`	betweenness x motorcyclist_casualty	Major junction x motorcycle collision
`speed_x_pedestrian`	speed_limit x pedestrian_casualty	Higher speed amplifies pedestrian injury
`hour_x_work`	hour x work_journey	Rush-hour commuter collision pattern
`runoff_crash`	left_carriageway AND skid_overturn	Combined loss-of-control flag
`btwn_x_pedestrian`	betweenness x pedestrian_casualty	Major junction x pedestrian collision

Three feature configurations were evaluated:

Variant	Features	Thresh-old	AUC	MCC	Sensitivity	Specificity	G-Mean
(1) Original – graph	40	0.48	0.7193	0.2422	0.5305	0.7674	0.6381
(2) Original + 7 interactions	50	0.42	0.7333	0.2538	0.6394	0.6942	0.6662
(3) Original + 7 interactions, pruned	45	0.43	0.7362	0.2583	0.6249	0.7110	0.6666

- Removing graph features (variant 1) reduced AUC by 0.0137 and sensitivity by 0.109 relative to variant 2, confirming centrality measures carried genuine signal.
- Adding seven interaction terms (variant 2) improved AUC to 0.7330 and raised sensitivity to 0.639.
- Pruning five zero/negative-permutation-importance features then yielded the strongest overall configuration (variant 3) with better specificity balance (0.711) than variant 2.

```
# Three-way ablation comparing:
#   (1) Original minus graph     — drop degree/betweenness/closeness centrality
#   (2) Original plus interactions — add 7 confirmed interaction terms
#   (3) Original + interactions, pruned — drop 5 negative-permutation features
#
# Drop candidates (perm importance ≤ 0):
#   weather_conditions, ped_jaywalking, child_casualty, quarter, ped_on_crossing

import numpy as np
import pandas as pd
from xgboost import XGBClassifier
from imblearn.over_sampling import SMOTE
from sklearn.model_selection import train_test_split
from sklearn.metrics import matthews_corrcoef, roc_auc_score, confusion_matrix

#    Confirmed interaction terms
INTERACTION_TERMS = [
    ('hour_x_light',        'hour_of_collision',      'light_conditions'),
    ('imd_x_pedestrian',    'mean_casualty_imd',      'pedestrian_casualty'),
    ('btwn_x_motorcyclist', 'betweenness_centrality', 'motorcyclist_casualty'),
    ('speed_x_pedestrian',  'speed_limit',            'pedestrian_casualty'),
    ('hour_x_work',         'hour_of_collision',      'work_journey'),
    ('btwn_x_pedestrian',   'betweenness_centrality', 'pedestrian_casualty'),
]
```

```
GRAPH_FEATURES  = ['degree_centrality', 'betweenness_centrality', 'closeness_centrality']
PRUNE_FEATURES  = ['weather_conditions', 'ped_jaywalking', 'child_casualty',
                   'quarter', 'ped_on_crossing']

#    Build interaction columns on the original feature set
features_original = feature_cols.copy()

valid_interactions = []
for name, feat_a, feat_b in INTERACTION_TERMS:
    if feat_a in features_original and feat_b in features_original:
        model_df[name] = model_df[feat_a] * model_df[feat_b]
        valid_interactions.append(name)
    else:
        print(f'  Warning: {name} skipped — {feat_a} or {feat_b} not in feature set')

if 'any_left_carriageway' in features_original and 'any_skid_overturn' in features_original:
    model_df['runoff_crash'] = (
        (model_df['any_left_carriageway'] == 1) &
        (model_df['any_skid_overturn'] == 1)
    ).astype(int)
    valid_interactions.append('runoff_crash')

features_with_inter   = features_original + valid_interactions
features_no_graph     = [f for f in features_original if f not in GRAPH_FEATURES]
features_pruned       = [f for f in features_with_inter if f not in PRUNE_FEATURES]

print(f'Feature set sizes:')
print(f'  Orig.:                        {len(features_original):>3} features')
print(f'  Orig. — graph:                {len(features_no_graph):>3} features  '
      f'(dropped: {GRAPH_FEATURES})')
print(f'  Orig. + {len(valid_interactions)} interactions:            '
      f'{len(features_with_inter):>3} features')
print(f'  Orig. + interactions, pruned:  {len(features_pruned):>3} features  '
      f'(dropped: {PRUNE_FEATURES})')

#    XGBoost best params from CV
if 'cv_results' in dir():
    clean_params = {k.replace('model__', ''): v
                    for k, v in cv_results['XGBoost']['best_params'][-1].items()}
else:
    clean_params = {
        'learning_rate': 0.1, 'max_depth': 4, 'n_estimators': 500,
        'subsample': 0.8, 'scale_pos_weight': 3,
    }

#    Evaluate all three variants
variants = [
(f'1. Orig. — graph ({len(features_no_graph)} features)',              features_no_graph),
(f'2. Orig. + {len(valid_interactions)} interactions. ({len(features_with_inter)} features)', features_with_inter),
(f'3. Orig. + interactions., pruned ({len(features_pruned)} features)',  features_pruned),
]

results_ablation = []
for label, cols in variants:
    cols_avail = [c for c in cols if c in model_df.columns]
    X_abl = model_df[cols_avail]
    y_abl = model_df['severity_binary']

    X_tr, X_te, y_tr, y_te = train_test_split(
        X_abl, y_abl, test_size=0.3, random_state=42, stratify=y_abl
    )

    sm = SMOTE(random_state=42)
    X_tr_sm, y_tr_sm = sm.fit_resample(X_tr, y_tr)

    model_abl = XGBClassifier(
        random_state=42, eval_metric='logloss',
        verbosity=0, n_jobs=-1, tree_method='hist',
        **clean_params
    )
    model_abl.fit(X_tr_sm, y_tr_sm)

    prob = model_abl.predict_proba(X_te)[:, 1]
```

```python
    best_t, best_mcc_val = 0.5, 0.0
    for t in np.arange(0.05, 0.95, 0.01):
        pred_t = (prob >= t).astype(int)
        m_t = matthews_corrcoef(y_te, pred_t)
        if m_t > best_mcc_val:
            best_mcc_val = m_t
            best_t = t

    pred  = (prob >= best_t).astype(int)
    auc   = roc_auc_score(y_te, prob)
    mcc   = matthews_corrcoef(y_te, pred)
    tn, fp, fn_c, tp_c = confusion_matrix(y_te, pred).ravel()
    sens  = tp_c / (tp_c + fn_c)
    spec  = tn  / (tn + fp)
    f1    = 2*tp_c / (2*tp_c + fp + fn_c)
    gmean = np.sqrt(sens * spec)

    results_ablation.append({
        'Variant':     label,
        'Threshold':   round(best_t, 2),
        'AUC':         round(auc,   4),
        'MCC':         round(mcc,   4),
        'F1':          round(f1,    4),
        'G-Mean':      round(gmean, 4),
        'Sensitivity': round(sens,  4),
        'Specificity': round(spec,  4),
    })

    print(f'\n{label}')
    print(f'  Threshold={best_t:.2f}  AUC={auc:.4f}  MCC={mcc:.4f}  '
          f'Sens={sens:.4f}  Spec={spec:.4f}  G-Mean={gmean:.4f}  F1={f1:.4f}')
    print(f'  TN={tn:,}  FP={fp:,}  FN={fn_c:,}  TP={tp_c:,}')

#    Summary
print(f'\n{"="*70}')
print('  ABLATION SUMMARY')
print(f'{"="*70}')
ablation_df = pd.DataFrame(results_ablation)
print(ablation_df[['Variant','Threshold','AUC','MCC',
                   'Sensitivity','Specificity','G-Mean']].to_string(index=False))

base = results_ablation[1]  # interactions model as reference
print(f'\nDelta vs (2) Original + interactions:')
for r in results_ablation:
    if r['Variant'] == base['Variant']:
        continue
    print(f'  {r["Variant"]}')
    print(f'    ΔMCC={r["MCC"]-base["MCC"]:+.4f}  '
          f'ΔSens={r["Sensitivity"]-base["Sensitivity"]:+.4f}  '
          f'ΔAUC={r["AUC"]-base["AUC"]:+.4f}')
```

```
Feature set sizes:
  Orig.:                          43 features
  Orig. − graph:                  40 features  (dropped: ['degree_centrality', 'betweenness_centrality',
  ↪  'closeness_centrality'])
  Orig. + 7 interactions:             50 features
  Orig. + interactions, pruned:   45 features  (dropped: ['weather_conditions', 'ped_jaywalking', 'child_casualty',
  ↪  'quarter', 'ped_on_crossing'])

1. Orig. − graph (40 features)
  Threshold=0.48  AUC=0.7193  MCC=0.2422  Sens=0.5305  Spec=0.7674  G-Mean=0.6381  F1=0.3832
  TN=20,107  FP=6,093  FN=2,311  TP=2,611

2. Orig. + 7 interactions. (50 features)
  Threshold=0.42  AUC=0.7330  MCC=0.2538  Sens=0.6394  Spec=0.6942  G-Mean=0.6662  F1=0.3914
  TN=18,188  FP=8,012  FN=1,775  TP=3,147

3. Orig. + interactions., pruned (45 features)
  Threshold=0.43  AUC=0.7362  MCC=0.2583  Sens=0.6249  Spec=0.7110  G-Mean=0.6666  F1=0.3951
  TN=18,627  FP=7,573  FN=1,846  TP=3,076

======================================================================
  ABLATION SUMMARY
======================================================================
                                     Variant  Threshold    AUC    MCC  Sensitivity  Specificity  G-Mean
                1. Orig. − graph (40 features)       0.48 0.7193 0.2422       0.5305       0.7674  0.6381
```

```
      2. Orig. + 7 interactions. (50 features)       0.42 0.7330 0.2538       0.6394       0.6942  0.6662
3. Orig. + interactions., pruned (45 features)       0.43 0.7362 0.2583       0.6249       0.7110  0.6666

Delta vs (2) Original + interactions:
  1. Orig. − graph (40 features)
    ΔMCC=-0.0116  ΔSens=-0.1089  ΔAUC=-0.0137
  3. Orig. + interactions., pruned (45 features)
    ΔMCC=+0.0045  ΔSens=-0.0145  ΔAUC=+0.0032
```

---

7. Discussion

7.1 Model Insights

Model Performance. XGBoost proved the most robust classifier (AUC 0.724 ± 0.008). By implementing scale_pos_weight=3, the model successfully prioritized the minority class during training, achieving a sensitivity of 0.474. Applying an MCC-optimal threshold of 0.43 further boosted sensitivity to 0.625, proving that internal weighting combined with modest threshold tuning is superior to post-hoc adjustments alone.

Network centrality. The graph ablation (variant 1) quantified the independent contribution of network topology: removing degree, betweenness, and closeness centrality reduced AUC by 0.014 and sensitivity by 0.109 (ΔMCC = −0.012), confirming that junction topology carries genuine signal beyond casualty type and road environment, consistent with Sarkar et al. (2018).

SHAP interpretability. Deprivation features dominated SHAP importance rankings with imd_decile ranked first and mean_casualty_imd ranked second, capturing area and individual deprivation channels respectively. pedestrian_casualty ranked first in permutation importance (ΔMCC = 0.085), confirming pedestrian involvement as the single strongest univariate KSI predictor. motorcyclist_casualty returned positive permutation importance (ΔMCC = 0.026, ranked sixth) reflecting higher risks faced by motorcylists. any_hit_offroad_object generated negative SHAP values, consistent with predominantly low-speed, single-vehicle incidents against street furniture.

Interaction terms and pruning Dropping five zero-negative permutation importance variables in variant 3 improved AUC by 0.003 and MCC by 0.005 over variant 2. Adding seven interaction terms contributed a 0.014 AUC lift, successfully capturing compound risk mechanisms that constituent features could not represent.

7.2 Limitations

STATS19 is a police-reported dataset subject to systematic under-recording of slight injuries, which introduces class-boundary noise. The absence of key contributory factors such as impact speed, seatbelt usage, alcohol/drug impairment also limits the model's ability to capture behavioural risk, which likely explains a substantial portion of residual prediction error.

Network centrality was computed on a static 2023 OSMnx snapshot, and the model is cross-sectional, unable to capture temporal trends in network risk.

SMOTE generates synthetic minority-class observations that may not reflect realistic collision profiles, though its use within the cross-validation pipeline prevents data leakage.

The near-flat cross-deprivation KSI rate gradient (18.8-21.2% from most to least deprived) in the exploratory data analysis recasts London's inequality in road safety as an exposure disparity rather than a severity disparity since deprived areas experience 1.64x more collisions, but not materially worse outcomes per collision.

---

8. Conclusion

This study demonstrated that integrating graph-derived network centrality measures with casualty characteristics and deprivation indicators **meaningfully improves KSI prediction** for London road collisions. Ablation affirmed that graph centrality provides a distinct signal, confirming that network topology helps to capture residual risk beyond casualty type or road environment.

The optimized XGBoost configuration utilizing targeted interaction terms achieved an **AUC of 0.736** and an **MCC of 0.258**. By correctly identifying **close to 2 in 3 (62.5%)** of KSI incidents, the model offers a minimally viable tool that can sharpen safety intervention. Beyond predictive power, SHAP analysis uncovered a "deprivation paradox". While per-incident KSI rates remain flat across deprivation deciles, higher collision frequency in deprived areas drives systemic per-population inequality.

Future refinements should prioritize the integration of temporal speed, graph and flow data to better capture the evolving nature of urban risk.

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